Microeconomics
Business 33801

Professor Lars Stole
AXP-21, Autumn 2020

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Print your name here.

Mid-Term Examination, Business 33801: (Friday, 27 November 2020)

Read all questions carefully and write answers in the space provided. If you run out of space, you may write on the back of the exam pages; if you run out of space on the back of the exam, you may add additional pages of paper. Graphs must be thoroughly labeled for full credit. Show your work and explain all equations on numeric questions. Be concise – irrelevant remarks will not count for anything. Be careful to budget your time between various parts, taking into account the points allocated to each question (the number in parentheses); these points vary depending upon the question.

There are 9 questions with a total of 90 points possible. You have 90 minutes for the examination, so assign about one minute per point. Allocate your time wisely.

YOU MUST READ AND SIGN THE HONOR STATEMENT BELOW IN ORDER TO RECEIVE CREDIT FOR THE EXAMINATION.

> I pledge my honor that I have not violated the Honor Code during this examination. I will not communicate any information related to this examination to *any* individual that has not already seen the examination until after 4 December 2020.

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Sign your name here.

True-False-Uncertain (42 points total)

Indicate whether the italicized portion of each statement is True, False, or Uncertain and EXPLAIN your answer. Read each question carefully. You may take the non-italicized portion of the statement as true. **No credit will be given for answers without explanation.**

Question 1 (6 points) True, False, Uncertain? [Explain your answer.] The price of quinoa (an edible grain) has recently risen due to an unexpected, but permanent, increase in demand. Nothing else changes in the economy. *The change in quinoa demand will cause the short-run supply curve for quinoa to shift outwards over time in response to the higher quinoa prices, but will have no effect on the long-run supply curve of quinoa.*

Question 2 (8 points) True, False, Uncertain? [Explain your answer.] Prue sells Cornish pasties in a perfectly competitive market at the price of $p = £2$ per pasty. (A pasty is a small meat pie.) Prue chooses to bake and sell $q^* > 0$ pasties each day, such that the marginal output rule for short-run profit maximization is satisfied. At this output, her average short-run opportunity cost is less than the market price.

(a). *Prue's short-run marginal cost at q^* is less than 2.*

(b). *Prue's costs exhibit short-run economies of scale at q^*.*

Question 3 (8 points) True, False, Uncertain? [Explain your answer.] Noel currently buys 20 cups of coffee per month at a price of £2 per cup. For reasons independent of supply or demand, the government has decided to subsidize coffee sales with the net effect that the price Noel pays for coffee falls to £1.50 per cup. In order to fund the subsidy, the government has decided to charge everyone (including Noel) an additional £120 per year in taxes (i.e., £10 per month). *This policy has made Noel better off and he will drink more coffee.*

Question 4 (8 points) True, False, Uncertain? [Explain your answer.] Paul has preferences that are represented by expected utility. Paul is offered a choice between a certain payment of 80 and a gamble that pays 400 with probability $\phi_1 = 0.75$ and generates a loss of 800 with probability $\phi_2 = 0.25$. Paul strictly prefers the gamble over the sure thing.

(a). *Paul is risk averse.*

(b). *Paul's risk premium of the gamble is less than 20.*

Question 5 (6 points) True, False, Uncertain? [Explain your answer.] Suppose that $MP_L = 40$ and $MP_K = 60$. *If $w = 60$ and $r = 40$, then the firm could reduce costs in the long run by substituting from capital into labor.*

Question 6 (6 points) True, False, Uncertain? [Explain your answer.] Three months ago, Greenmill Coffee Stores purchased 5,000 pounds of coffee beans (to be sold in Greenmill's coffee shops) at the Chicago Board of Trade Commodity Exchange. In the last three months, the price of coffee beans has risen unexpectedly by 20 percent. The coffee-shop industry in Chicago is a perfectly competitive industry. *In the short run, Greenmill has a lower opportunity cost of coffee beans relative to its competitors and should therefore charge a lower price than competing coffee shops.*

Short problems (48 points total)

Question 7 (8 points) More rain improves the yield of grapes; dryer weather reduces the yield. Hot temperatures improve the taste of grapes; cooler temperatures worsen the taste of grapes. Suppose that the demand for grapes is downward sloping, the supply is upward sloping, and the market is competitive.

(a). If weather is currently moderate (midway between dry and wet, and midway between hot and cool), what change(s) to the weather are certain to generate an increase in the quantity of grapes?

(b). If weather is currently moderate (midway between dry and wet, and midway between hot and cool), what change(s) to the weather are certain to generate an increase in the price of grapes?

Question 8 (20 points) Suppose that supply and demand are given by the equations $Q^s = 3 + p$ and $Q^d = 9 - 2p$ where quantity is measured in metric tons and price is measured in British pounds sterling (£) per metric ton.

(a). What is the equilibrium price and output sold?

(b). What is the point elasticity of demand at the equilibrium price and quantity?

(c). Assume that one British pound (£) is equal to two Singapore dollars (S$). If we convert the demand and supply equations to be in terms of Singapore dollars, how does the elasticity derived in part (b) change? Explain.

(d). Suppose that the supply curve shifts outward slightly due to improved production methods. Using your answer in (b), what happens to industry revenues. Explain.

Question 9 (20 points) Mel and Sue own a small firm (M&S distillers) which produces vodka in a competitive market. Their total opportunity cost of producing q cases of vodka is the same in the short run and the long run and is given by

$$C(q) = \begin{cases} 64 + q^2 & \text{if } q > 0 \\ 0 & \text{if } q = 0. \end{cases}$$

(a). M&S's average opportunity cost curve is U-shaped. What is the critical price at which M&S will be indifferent to producing vodka and shutting down?

(b). Suppose that the production of vodka is a constant-cost industry and the cost function above represents long-run opportunity costs for all firms in the industry. Assuming the market operates, what will be the long-run equilibrium price?

(c). Suppose that the industry is not constant cost, and M&S can sell as much vodka as they like at a market price of $p^* = 18$. How much should they sell? [Consider both rules of profit maximization for full credit.]

(d). In addition to selling as much as they want at $p = 18$, M&S also has the opportunity to sell up to 10 specially-labelled cases each day to a retailer who wants their own custom label for a higher price of $p = 24$. Producing a case with different brand labels does not cost any more than producing a standard unit. The total cost of producing q_s standard cases and q_c custom cases is $C(q_s + q_c)$, where the total opportunity cost function is the same as above. Given the opportunity to produce for both the competitive market at $p = 18$ and up to 10 units at $p = 24$, how much should M&S produce in total?

Microeconomics, Business 33801 **Professor Lars Stole**

Answers to mid-term examination

1 True. In the long run, more quinoa will be grown given the higher market price for quinoa. This implies that the short-run supply curve shifts out as more "entry" occurs (i.e., more acres of quinoa planted). The long-run supply curve already captures these expansion effects, so the long-run supply curve should not shift without additional exogenous variables impacting the supply side of the market.

2 (a). False. The marginal output rule requires $MR = MC_{sr}(q^*)$. Because $MR = p$, we have $MC_{sr}(q^*) = 2$.

(b). False. Because $AC^O_{sr}(q^*) < p = 2 = MC_{sr}(q^*)$, average cost is below marginal cost and so it must be increasing. Hence, there are diseconomies of scale.

3 True. There are two points to note. First, if Noel continues to consume the same number of cups of coffee as before, he can continue to purchase the same amount of other goods as well. Thus, he cannot be worse off from this policy. Indeed, because the relative price of coffee to other goods has fallen, consuming 20 cups per week is likely to be suboptimal and he will do strictly better by drinking more coffee. Thus, he would be better off than before. If you said "Uncertain", then you must note that the *only* possibility for Noel to be indifferent is if the change in relative prices is not enough to cause him to buy more than 20 cups of coffee per month, and that otherwise he would be strictly better off.

4 (a). Uncertain. Because Paul prefers a risky portfolio with higher expected payoff to a lower no-risk payoff, we cannot tell whether or not Paul is risk averse. He may be only mildly risk averse and the extra payoff from the risky portfolio more than compensates him for the risk; he could also be risk loving.

(b). True. The expected payoff of the gamble is $E[\tilde{x}] = \frac{3}{4}(400) + \frac{1}{4}(-800) = 100$ which exceeds the sure thing of 80. Because Paul strictly prefers the risky asset with 20 dollars more in expected return to the sure thing, the risk necessarily imposes a cost on him that is less than 20 dollars: i.e., $\rho < 20$.

5 False. The firm will want to substitute from labor into capital to reduce costs. One way to see this is to note that the output produced by an additional dollar spent on labor is $\frac{MP_L}{w} = \frac{40}{60} = 0.67$, while the output produced by an additional dollar spent on capital is $\frac{MP_K}{r} = \frac{60}{40} = 1.5$. Thus, reallocating inputs toward capital and away from labor will reduce costs of production.

Note: You could have also noted that such a reallocation from capital to labor may be impossible because the isoquants are kinked (i.e., the inputs are perfect complements) or the firm is at a boundary solution, but you did not need to depart from a discussion of the regular case to obtain full marks.

6 False. Greenmill obtained a capital gain for its purchases of coffee, but its opportunity cost of coffee is the value of the beans on the commodity market. Hence, its opportunity cost for beans has risen by the same 20 percent as its competitors.

7 (a). Equilibrium quantities will increase if demand and supply both shift outward. Rain shifts supply outward; heat shifts demand outward. Thus, rain plus heat is certain to increase the quantity of grapes produced and consumed.

(b). Equilibrium prices will increase if demand shifts outward and supply shifts inward. Dry weather shifts supply inward; heat shifts demand outward. Thus, dry and hot weather is certain to increase the price of grapes.

8 (a). $Q^s = Q^d$ implies that $3 + p = 9 - 2p$ so $p^* = 2$ £/metric ton. At this price, $Q^* = 5$ metric tons.

(b). The elasticity of demand is

$$\varepsilon^d = \frac{dQ^d}{dp}\frac{p}{Q} = (-2)\frac{2}{5} = -0.8.$$

(c). The elasticity doesn't change. Elasticities are independent of the units of measurement used for prices and quantities.

(d). Given the answer in (b), we are in the inelastic region. If supply shifts outward slightly, the equilibrium point moves down the demand curve. Because we are in the inelastic region, this implies that industry revenues must decrease.

9 (a). The minimum of the average cost curve occurs at the point of intersection with the marginal cost curve. Thus,

$$2\hat{q} = \frac{64}{\hat{q}} + \hat{q},$$

or $\hat{q} = 8$. At this point, average cost is $AC(\hat{q}) = \frac{64}{8} + 8 = 16$.

(b). The long-run supply curve in a constant-cost industry is horizontal at minimum average cost. Given the answer in (a), we know the supply curve is horizontal at $p = 16$. Thus, if vodka is sold in equilibrium, it must be at a price of $p = 16$ for any demand curve.

(c). Because marginal revenue is the market price in a competitive market, we have $MR = p = 18$. The marginal output rule requires at

$$MC(q) = MR = 18,$$

and so $2q = 18$ which implies $q^* = 9$ (assuming it is optimal to produce). We also need to check the shutdown rule to be complete in our answer. Given minimum average cost is 16 and $p = 18 > 16$, we know that average revenue exceeds average cost, and hence economic profit is positive. [You could also have computed economic profit directly. $\pi = p * q - C(q) = 18(9) - 64 - (9)^2 = 162 - 64 - 81 = 17 > 0$.]

(d). The key to answering this problem is to note that the marginal revenue for the first 10 units is $MR = 24$ because you would always sell those items to the custom vendor. For $q > 10$, however, marginal revenue falls to $MR = 18$, while the marginal cost of the next unit is $MC(10) = 2(10) = 20 > 18$. Thus, it does not make sense to produce beyond the custom order: $q_c^* = 10$ and $q_s^* = 0$.